

UDP - User Datagram Protocol

Module 13

Duration: 22:53

Video: https://www.youtube.com/watch?v=g4p_kJxiRHk

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agriculture University, Ludhiana

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

Hello friends I am engineer Sunil Kumar from school of electrical engineering and information technology Punjab Agriculture University Ludhiana I welcome you all to the video lecture series of cyber and information security in this lecture we will study about UDP first of all please see the contents which I shall cover after covering brief introduction about UDP I shall take up characteristics of user datagram which will be followed by UDP services in the last I shall take up UDP features and UDP packages so now let us start UDP is a simple transfer protocol that extends the host to host delivery of packets of the underlying Network into a process to process communication since there are many processes running on a given host UDP needs to add a level of Dem

multiplexing which allowing multiple application processes on each host to share the network figure shows the relationship of the user datagram protocol to the other protocols and layers of the tcpip protocol suit UDP is located between the application layer and the IP layer and serve as the intermediary between the application programs and the network operations a transport layer protocol usually has several responsibilities one is to create a process to process communication UDP uses port number to accomplish this another responsibility is to provide control mechanism at the transport layer UDP does this task at a very min minimal level there is no flow control mechanism and there is no acknowledgement for received packets UDP does provide error control to some

extent if UDP detects an error in the received packet it silently drop it UDP is a connectionless unreliable transfer protocol it does not add anything to the services of Ip except for providing process to process communication instead of host to host communication UDP is a very simple protocol using a minimum of overhead if a processor wants to send a small message and does not care about reliability it can use UDP sending a small message using UDP takes much less interaction between the sender and the receiver than TCP next user datagram UDP packets called user datagrams have a fixed size header of 8 bytes figure shows the form of the user datagram the fields are as follows First Source port number this is

the port number used by the process running on the source host it is 16bit long which means that P number can range from 0 to 65,535 if the source host is a client the port number in most cases is an inal port number requested by the process and chosen by the UDP software running on the source host in the source host is the server the p number in most cases is a well-known port number next destination port number this is the port number used by the process running on the destination host it is also 16 bits long if the destination host is the server the port number in most cases is a well-known port number if the destination host is the client the port number in most cases and inferable port number in this case the server copies the inferable port number

it has received in the request packet this is a 16bit field that defines the total length of the user datagram and header Plus data the 16 bits can define a total length of 0 to 65,535 bytes however the total length needs to be much less because a UDP user datagram is stored in a and IP datagram with the total length of 65,535 bytes the length field is a UDP user datagram is actually not necessary a user datagram is encapsulated in an IP datagram there is a field in IP datagram that defines the total length there is another field in the IP datagram that defines the length of the header so if we subtract the value of the second field from the first we can deduce the length of the user datagram that is

encapsulated in an IP datagram UTP length equal to IP length minus IP headers length however the designers of the UDP protocol felt that it was more efficient for the destination UDP to calculate the length of the data from the information provided in the UDP used datagram rather than ask the IP software to supply this information we should remember that when the IP software delivers the UDP user data gram to the UDP layer it has already dro the IP headers next checkm this field is used to detect errors over the entire user datagram that is header Plus data I will discuss the dump example of UDP header in hexadecimal format the UDP header is CB 8400 d 0 01 C 0 01 C some questions based on this UDP header first question

what is a source port number The Source port number is the first four hexadecimal digits like C B 846 which means that the source Board number is 5210 next question what is a destination port number the answer is that the destination port number is the second four hexadecimal digits that are 0 0 0 D14 which means that the destination port number is 13 next question what is the total length of the user datagram the third four hexadecimal digits 0 0 1 c16 defines the length of the whole UDP packets as 28 bytes next question what is the length of data the length of the data is the length of the whole packet minus the length of the header means $28 - 8$ equal to 20 bytes next question is the packet directed from a client to a server or visce Versa answer since the destination port number is 13 the packet

is from the client to the server next question what is the client process the client process is the daytime next UDP Services I discuss the general Services provided by the UDP in transport layer first process to process communication UDP provides process to process communication

using sockets with the combination of IP addresses and port numbers several port numbers used by UDP are as follows port 5 is for Echo Port 7 is for discard port 9 is for users port 11 is for daytime Port 13 is for COD Port 17 is for charging port 19 is for domain Port 20 is for boot PS Port 21 is for boot PC Port 22 is for tftp Port 23 is for RPC Port 24 is for ntp Port 25 is for SNMP Port 26 is for SNMP trap next service is connectionless service UDP provides a connectionless service this means that each user

datagram sent by UDP is an independent datagram there is no relationship between the different users datagram even if they are coming from the same Source process and going to the same destination program the user datagrams are not numbered also there is no connection establishment and no connection termination as is the case of TCP this means that each user datagram can travel on a different path one of the ramifications of being connectionless is that the processes that uses UDP cannot send a stream of data to UDP and expect UDP to chop them into different related users datagrams instead each request must be small enough to fit into one user datagram in the case of UDP only those processes

sending short messages when messages less than 65535 bytes next flow control UDP is a very simple protocol there is no flow control and hence no window mechanism the receiver may overflow with incoming messages the lack of flow control means that the process using UDP should provide for this service if it is needed next error control there is no error control mechanism in a UDP except for the check sum this means that the sender does not know if a message has been lost or duplicated when the receiver detects an error through the check sum the user datagram is silently discarded the lack of error control means that the process using UDP should provide for this service if it is needed next checksum UDP check sum calculation is different

from the 104 IP here the checksum includes three sections a pseudo header a UDP header and the data coming from the application layer the pseudo header is the part of the header of the IP packet in which the user datagram is to be encapsulated with some fields filled with zeros if the checksum does not include pseudo header a user datagram may arrive safe and sound however if the IP header is corrupted it may be delivered to the wrong host the protocol field is added to ensure that the packets belong to UDP and to TCP we will see later that if a process can use either UDP or TCP the destination port number can be the same the value of the protocol field for UDP is 17 if this value is changed during

the transmission the checksum calculation at the receiver will detect and UDP drops the packet it is not delivered to the wrong protocol note the similarities between the pseudo header fields and the last 12 bytes of the IP address next congestion control since UDP is a connectionless protocol it does not provide congestion control UDP assumes that the packets sent are small and cannot create congestion in the network this assumption may or may not be true today when UDP is used for realtime transfer of audio and video next encapsulation and D encapsulation to send a message from one process to another process the UDP protocol

encapsulates and de encapsulates the messages first encapsulation when a process has a message to send through

UDP it means the message to UDP along with the pair of socket addresses and the length of data UDP receives the data and adds the UDP header UDP then passes the user datagram to IP with the socket addresses IP adds its own head using the value 17 in the protocol field which indicating that the data has come from the UDP protocol the IP datagram is then passed to the data link layer the data link layer receives the IP datagram which adds its own header and passes it to the physical layer the physical layer encodes the bits into electrical or Optical signals and send it to the remote machine next decapsulation when the message arrives at the destination host the physical layer decodes the signal into bits and passes

it to the data link layer the data link layer uses the header to check the data if there is no error the header and the trailer are dropped and the datagram is passed to IP the IP software does its own checking if there is no error the header is dropped and the user datagram is passed to UDP with the sender and the receiver IP addresses UDP uses the check sum to check the entire user datagram if there is no error the header is dropped and the application data along with the sender socket addresses is passed to the processes the sender socket address is passed to the processes in case it needs to respond to the message received next queing in UDP qes are associated with ports at the client side when a process

starts it request a port number from the operating system some implementations create both an incoming and outgoing cues associated with each process other implementations create only an incoming queue associated with each process note that even if a process wants to communicate with multiple processes it obtains only one p number and eventually one outgoing and one incoming queue the cues opened by the clients are in most cases identified by eeral P numbers the cues function as long as the process is running when the process terminates the cues are destroyed the client processes can send messages to the outgoing cues by using Source Board number specified in the request UDP removes the messages

one by one and after adding the UDP header deliver them to IP an outgoing queue can overflow if this happens the operating system can ask the client processes to wait before sending any more messages next multiplexing and Dem multiplexing in a host running a tcpip protocol suit there is only one UDP but possibly several processes that may want to use this service of UDP to handle this situation UDP multiplexes and D multiplexes first mul Lexing at the center side there may be several processes that need to send user datagrams however there is only one UDP this is a many to one relationship and requires multiplexing UDP accepts messages from different processes differentiated by their assigned P

numbers after adding the header UDP passes the user datagram to IP next d multiplexing at the receiver side there is only one UDP however we may have many processes that canes user datagrams this is one to many relationship and requires Dem multiplexing UDP receives user

datagrams from IP after error checking and dropping of the header UDP delivers each message to the appropriate process based on the port numbers next comparison between UDP and generic simple protocols we can compare UDP with the connectionless simple protocol the only difference is that the UDP provides an optional check sum to detect corrupted packets at the receiver side if the check sum is added to the packet the receiving UDP can check the packet and discard the packet if it is corrupted no feedback however is sent to

the center UDP is an example of the connectionless simple protocol with the exception of an optional checkm added to 17 minutes, 1 second packets for error detection next UDP features we briefly discuss some features of UDP and their advantages and disadvantages First connectionless Service as we mentioned previously UDP is a connectionless protocol each UDP packet is independent from other packets sent by the same application program this feature can be considered as an advantage or disadvantage depending on the application requirement it is an advantage if for example a client application needs to send a short request to a server and to receive a short response if the request and response can each fit in one single user datagram a connectionless service may be

preferable the overhead to establish and close the connection may be significant in this case in the connection oriented service to achieve the above goal at least nine packets are exchanged between the client and the server in connectionless service only two packets are exchanged the connectionless service provides less delay and the connection oriented service creates more delay if delay is an important issue for the application the connection L service is preferred next lack of error control UDP does not provide error control it provides an unreliable service most applications accept reliable services from a transport lay protocol although a reliable service is desirable it may have some side effects that are not acceptable to some applications when a

transport layer provides reliable Services if a part of the message is lost or corrupted it needs to resend this means that the receiving transport layer cannot deliver that part of the application immediately there is an uneven delay between different parts of the message delivered to the application layer some applications by Nature do not even notice these uneven delays but for some they are very crucial next lack of congestion control UDP does not provide congestion control however UDP does not create additional traffic in an error prone Network TCP May resent a packet several times and thus contribute to the creation of the congestion therefore in some cases lack of error control in UDP can be considered an advantage when congestion is a big issue next UDP

package to show how UDP handles the sending and receiving of UD package we present a simple version of UDP package we can say that UDP package involve five components first control Block Table input q a control block module an input module and a output module figure shows the five components and their interactions first control Block Table in our package UDP has control Block Table to keep track of the open ports each entry in this table

has a minimum of four Fields the state which can be free or in use the process ID the port number and the corresponding Q number next next input cues our UDP package uses a set of input cues one for each process in this design we do not use output cues next control block module the control block module is

responsible for the management of the control block table when a process starts it asks for a port number from the operating system the operating system assigns well-known port numbers to servers and imperial port numbers to the client the processes passes the process ID and the port number to the control block module to create an entry in the table for the process the field for Q number has a value of zero note that we have not include a strategy to deal with the table that is full next input module the input module receives a user datagram from the IP it searches the control Block Table to find an entry having the same port number as this user datagram if the entry is found the module uses the information in the entry

to NQ the data if the entry is not found it generates an icmp message next output module the output module is responsible for creating and sending user datagrams so friends this was all about UDP hope the concepts explained in this 22 minutes, 1 second lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank

you for